Thermodynamics and Statistical Physics

6th Exercise Class

Helmholtz free energy and other thermodynamic potentials; Thermodynamic equilibrium; Maxwell relations

Exercises for the Class

1. The so called natural variables of the internal energy are the entropy (S) and the volume (V): $U = U(S, V)$. For some processes the enthalpy is more useful:

   $$H = U + pV,$$

   where p is the pressure of the gas. What are the natural variables of the enthalpy?

2. The Helmholtz function of one mole of a certain gas is

   $$f = \frac{F}{n} = f_0(T) - \frac{a}{v} - RT \ln \frac{v - b}{C} + B,$$

   where $f_0(T)$ depends only on the temperature, v is the molar volume, and a, b, B and C are constant. Give the equation of state of the gas (relation among p, V and T).

3. Consider the Helmholtz free energy.

   (a) Derive the appropriate Maxwell relation.

   (b) Show that

   $$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p.$$

4. Derive the following equation

   $$C_V = -T\left(\frac{\partial^2 F}{\partial T^2}\right)_V.$$

5. Express in terms of measurable quantities the temperature change dT of a system when its volume changes by dV in an adiabatic quasi-static process.

6. Verify that

   $$C_p - C_V = \frac{TV\beta_p^2}{\kappa_T}.$$

7. The expansion of a gas through a small opening or a porous plug with the pressure on either side being maintained is called Joule-Kelvin expansion. The expansion is isenthalpic. The Joule–Kelvin coefficient is:

   $$\mu_{\mathrm{JK}} = \left(\frac{\partial T}{\partial p}\right)_H.$$

   Show that the coefficient can be expressed as

   $$\mu_{\mathrm{JK}} = \frac{1}{C_p}\left[T\left(\frac{\partial V}{\partial T}\right)_p - V\right].$$

8. Show that the chemical potential of an ideal gas at temperature T varies with pressure as

   $$\mu = k_{\mathrm{B}} T \ln\left(\frac{p}{p_0}\right) + \mu_0,$$

   where μ_0 is the value at reference point of pressure p_0 and temperature T. The gas consists of a single type of particle only.

Homework to Be Submitted

1. Give the difference of the the isothermal and the adiabatic compressibilities, $\kappa_T - \kappa_S$ with measurable quantities.

Homework for Practice

1. Using the equations for an ideal gas

   $$pV = Nk_{\mathrm{B}}T,$$
   $$U = \frac{f}{2}Nk_{\mathrm{B}}T$$

   determine

   (a) the internal energy $U = U(S, V, N)$ as a function of its natural variables;

   (b) the free energy $F = F(T, V, N)$ as a function of its natural variables.

2. Prove the

   $$\left(\frac{\partial \mu}{\partial p}\right)_{S,N} = \left(\frac{\partial V}{\partial N}\right)_{S,p}$$

   Maxwell relation.

3. A block of metal is subjected to an adiabatic and reversible increase of pressure from p_1 to p_2. Show that the initial and final temperatures T_1 and T_2 are related as

   $$\ln\left(\frac{T_2}{T_1}\right) = \frac{V\beta_p}{C_p}(p_2 - p_1).$$

   You may assume that the volume of the block stays approximately constant during the compression.